

provide alert about their presence. Drone monitoring exploits various features of flying drones, such as the heat emitted, sound generated, and radio frequency (RF) spectrum used.

Drone monitoring equipment can be passive—simply looking or listening, or active—sending a signal out and analysing what comes back. Drones have low radar cross-section (RCS), slow speed, and usually small size, making the task of detection difficult and thereafter identification and localisation even more difficult. Various technologies available include RF-analysers, radars, acoustic sensors, optical/infrared sensors, thermal detectors, and hybrid detectors.

Radio frequency (RF) analysers. Drone-detection RF monitoring system operators communicate with drones through RF signals, which include sensor outputs, location data, flight commands, and their media access control (MAC) address—a unique identification number. RF analysers intercept these signals to identify the make, model, and controller of the approaching drones. Some high-end analysers have the capability to triangulate the drones and their controllers using multiple receiver units spread far apart.

RF analysers provide long range, are low-cost, can detect and identify multiple drones and their controllers, and are passive. However, they are not capable of detecting drones that do not exchange RF signals, use different frequency bands, have autonomous navigation and/or unknown control protocols. RF analysers are less effective in crowded RF areas.

Radar. Though radar is a useful tool for detection of drones, there are limitations like low altitude, velocity of flying, and their small RCS, making it difficult to distinguish them from the noise and clutter present. Sometimes, multi-static radars are used to monitor, track, and analyse the energy back-scattered from rotating parts like propellers and rotors for drone detection.



An Amazon drone making a delivery (Source: <https://www.asme.org>)

Radars are classified as 2D and 3D by the type of the phase array antenna used. 2D radars use passive electronically scanned array antennas (PESAs) and provide relatively large detection range. However, 3D radars use active electronically scanned array antennas (AESAs), have relatively short detection range, can self-correct errors, and support wideband detection. 3D radars are preferred as they can estimate the altitude of target objects. Radar based detection systems offer longer range and constant observability compared with RF scanners, but have regulatory limitations for use.



A drone detection RF monitoring system (Source: <https://a8242685.en.made-in-china.com>)

At times radars find it difficult to distinguish between drones and the obstacles when the drones hover in one position or fly at low speed and are mostly employed in conjunction with other technologies (camera, RF scanner, etc). Radars emit high-power RF signals, therefore other facilities using radars, such as airports, may face RF interference issues. Partial spectral overlap between radar and radio waves can cause signal interference and poor performance of both radar and the radio networks.

Acoustic sensors (microphones).

During flight the sound generated by the rotors can be utilised in detection, classification, and localisation of drones. Acoustic sensors, usually microphones or microphone arrays, are used to detect the sound generated and calculate the direction of the flight using algorithms such as multiple signal classification (MUSIC).

Acoustic sensors can detect drones within the near-field and even those that are operating autonomously and not emitting RF radiations. Some systems use tetrahedron acoustic arrays (four microphones in a tetrahedral array located about one metre above the ground) for detection, recognition, and to find out the direction of arrival.

Acoustic sensors can detect drones in the ground clutter where other technologies can struggle, are a great gap-filler, provide high mobility, are quickly deployable, and are complete-